

Multi-Threshold Pass-Count Preservation under Bounded Score Perturbations

Research Architecture Runtime

operator/multi-threshold

2026-07-29

Abstract

For a finite threshold list, the pass-count $C(x) = \sum_i \mathbf{1}\{x \geq T_i\}$ is invariant under $|x' - x| \leq \varepsilon$ whenever every cut is buffered ($x \geq T_i + \varepsilon$ or $x < T_i - \varepsilon$). Lean `LEAN_FULLL`.

This theorem is: Derived

1 Problem

The multi-threshold operator returns the Nat pass-count $C(x; T_\bullet) = |\{i : x \geq T_i\}|$ over a finite list of cuts (equality passes).

2 Stability notion

Perturbation: $|x' - x| \leq \varepsilon$. Hypothesis `allCoordsStable`: each T_i satisfies $x \geq T_i + \varepsilon$ or $x < T_i - \varepsilon$. Stability: $C(x'; T_\bullet) = C(x; T_\bullet)$. Sharpness: if x lies in $[T - \varepsilon, T + \varepsilon)$ for a single cut, some admissible x' changes the count.

3 Definitions

Definition 1 (Pass bit / count). `passBit`(x, T) = 1 if $x \geq T$ else 0; $C(x; []) = 0$ and $C(x; T::\text{tail}) = \text{passBit}(x, T) + C(x; \text{tail})$.

Assumptions. $\varepsilon \geq 0$, $|x' - x| \leq \varepsilon$, and `allCoordsStable`.

4 Theorem

Theorem 2 (Count preservation). *If $|x' - x| \leq \varepsilon$ and every $T \in T_\bullet$ is buffered as above, then $C(x'; T_\bullet) = C(x; T_\bullet)$.*

Theorem 3 (Sharpness). *If $T - \varepsilon \leq x < T + \varepsilon$, there exists $|x' - x| \leq \varepsilon$ with $C(x'; [T]) \neq C(x; [T])$.*

5 Intuition

Each buffered cut is a scalar threshold bit; the count is the sum of those bits. If any cut sits in its unstable band, that bit can flip.

6 Examples

Example 4. $T = (1, 3, 5)$, $x = 4$, $\varepsilon = 0.2$: passes $T = 1, 3$ so $C = 2$; buffered cuts keep $C = 2$. If $T = 3$, $x = 3.1$, $\varepsilon = 0.2$, then $x' = x - \varepsilon$ can drop the pass bit.

7 Proof

Preservation. Induct on the threshold list. The empty list has count 0. For $T::\text{tail}$, `allCoordsStable` gives a scalar buffer on T , so threshold preservation yields $\text{passBit}(x', T) = \text{passBit}(x, T)$. The inductive hypothesis equates the tail counts. Adding proves Theorem ??.

Sharpness. Assume $T - \varepsilon \leq x < T + \varepsilon$. If $x \geq T$, take $x' = x - \varepsilon$ (bit drops). If $x < T$, take $x' = x + \varepsilon$ (bit rises). This proves Theorem ??.

8 Formal statement

Lean module

`Research.Operators.MultiThreshold.Preservation`

Source file

`lean/Research/Operators/MultiThreshold/Preservation.lean`

Theorems

- `multi\_threshold\_preservation`
- `multi\_threshold\_sharpness`

Note

Depends on `Research.Operators.Threshold.Preservation` per cut.

Certificate

`lean/certificates/multi-threshold/multi-threshold-count-preservation`

Status

LEAN_FULL on Mathlib $\mathbb{R}$ (REAL_MATHLIB)

9 Proof dependencies

- `Threshold.Preservation` (each pass bit).
- List induction on the threshold spine.
- No ranking / argmax theorems required.

10 Consequences

Unordered pass-counts for multiple cuts; constraint conjunction/disjunction reuse buffered bits. Ordered bucket indices are a separate ranking/bucket operator.